



Ezenia Diaz-Lembert

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About myself

Electrical engineering graduate with experience in audio consumer electronics, power system modeling, and user-focused device design. Motivated by building technology that enables intuitive, effective human-device interaction.

Work experience

Energy Control Center Designer - Power System Applications Analyst | Avangrid | 07/07/2025 - Current | Augusta, United States

- Designing and maintaining a transmission network model for Central Maine Power's energy management system.
- Conducting studies on smart grid impact via contingency analyses.
- Wrote a Python script for data synchronization across GIS, Siemens Spectrum Energy Management System, and Distribution One-Line Diagrams, reducing reconciliation time by 15 hours/month.

Global R&D Consumer Systems and Architecture Co-op | Bose Corporation | 06/01/2025 - 23/05/2025 | Framingham, United States

- Designed Simulink models for FPGA simulation and synthesis in a Linux environment.
- Designed biquad filter models to emulate headset fits for FPGA-in-the-loop applications in testing and automation.
- Integrated external high-speed converters into the FPGA design for real-time testing.
- Validated Simulink model FPGA deployment with an oscilloscope and waveform generator.

Electrical Engineering Intern | Eaton Corporation | 03/06/2024 - 30/08/2024 | Moon Township, United States

- Developed C code in IAR Tool to improve Total Harmonic Distortion computation accuracy from 7th to 39th current harmonics for the PCD-Air Circuit Breaker team, reducing error delta by 80%.

Electrical Engineering Intern | BAE Systems | 25/09/2023 - 02/01/2024 | Merrimack, United States

- Conducted circuit design and analysis of phase-locked loops.
- Conducted Monte Carlo simulations and PCB layout for circuits in Cadence.

Electrical Engineering Intern | Halo Braid | 31/07/2023 - 22/09/2023 | Cambridge, United States

- Developed a motor control system using an Arduino for an automatic hair braiding system for black hair.
- Leveraged encoders and mechanical rotation device parameters to optimize braiding pace, improving efficiency by 50%.

Summer Intern | United States Department of Energy Office of Technology Transitions | 09/05/2022 - 26/08/2022 | Berkeley, United States

- Conducted customer discovery interviews to define target users and priority use cases.
- Analyzed a \$3.26B market, including competitive landscape and growth opportunities.
- Built a full Business Model Canvas to structure value proposition, channels, and revenue models.
- Identified and engaged potential adopters and industry partners to develop a 4-year commercialization roadmap for a Sandia National Laboratories technology.

Strategy Summer Intern | Massachusetts Executive Office of Technology Services and Security (EOTSS) | 07/06/2021 - 30/07/2021 | Remote, United States

- Conducted market and technology research to collaborate on technology accessibility projects for the Massachusetts Department of Housing and Development and the Commonwealth's Peace Officer Standards and Training Commission.

Corporate Liaison Officer | Harvard SEAS Society of Hispanic Professional Engineers | 14/09/2020 - 07/12/2023 |

Cambridge, United States

- Organized accessible events and group initiatives for the Harvard SHPE chapter, including Latines in STEM talks and Problem Set nights.
- Managed communications with corporate partners such as Jane Street.

Social Chair/Alumni Coordinator | The Harvard-Radcliffe Veritones | 14/11/2021 - 12/11/2023 | Cambridge, United States

- Organized events such as a 35th anniversary concert for more than 75 alumni and members of the a cappella group.
- Won first place at the International Championship of Collegiate A Cappella 2023 Quarterfinals competition.

Team Leader | Harvard First Year Retreat Experience (FYRE) | 15/08/2021 - 11/11/2021 | Cambridge, United States

- Introduced 12 incoming first-generation/low-income students to Harvard's academic and support resources as they transitioned from secondary school to college.
- Led group activities over a 4-month period.

Education & Training

Bachelor of Science in Electrical Engineering | Harvard College | 02/09/2020 - 11/02/2025 | Cambridge, United States

Field(s) of study:Electrical Engineering | **Final grade:** 3.549 | **Thesis:** "SoniqEQ: Dynamic Music Equalization Algorithm"

Language Skills

Mother tongue(s): **English** | **Spanish**

Skills

HDL(SystemVerilog,Vhdl, Verilog) | Python (computer programming) | Circuit design and simulation: LTspice | MATLAB | Simulink | Vivado | GIS

Creative Works

MIPS Multicycle Processor | 16/01/2023 - 08/05/2023

- Designed and implemented a datapath and control unit for a multicycle processor in SystemVerilog.
- Tested stimulation, created testbenches, and synthesized onto an FPGA.

SonicEQ: Dynamic Music Equalization Algorithm | 01/01/2024 - 01/12/2024

- Designed a real-time adaptive filter for personalized music equalization as a senior thesis project.
- Used a Convolutional Neural Network and spectral feature extraction on input audio signals for dynamic equalization.

Camera Pixel Array | 22/04/2024 - 13/05/2024

- Created a schematic, layout, and symbol for a pixel using Cadence Virtuoso.
- Passed LVD and DRC checks and stimulated pixel using Cadence Virtuoso.
- Tiled pixels to form a 640x480 pixel array.

RF Harvesting Model for Pulse Oximeter | 09/04/2024 - 10/05/2024

- Designed an RF energy harvesting model for a low-power pulse oximeter.
- Characterized antenna size, distance, and optimization for effective powering from different RF sources in urban environments.

Honours and Awards

Electrical Engineering 2023 Community Building MVP Award | Harvard University SEAS | 30/03/2023

- Recognized by the Harvard School of Engineering and Applied Sciences Undergraduate Electrical Engineering Department for organizing community events such as Women in EE Luncheons to help promote engagement and collaboration.

Hastings Award for Musical Performance | Harvard College Pforzheimer House | 01/05/2024

- Recognized by the Pforzheimer House Committee at Harvard College for musical performance activities in The Harvard-Radcliffe Veritones and winning first place at the International Championship of Collegiate A Cappella 2023 quarterfinals competition.